

Unit 5 – Jaccard Distance Calculations Exercise

Task:

The table shows the pathological test results for three individuals.

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	A
Mary	F	Y	N	P	A	P	N
Jim	M	Y	P	N	N	N	A

Calculate Jaccard distance/dissimilarity for the following pairs:

- (Jack, Mary)
- (Jack, Jim)
- (Jim, Mary)

(Jack, Mary) = 0.33

(Jack, Jim) = 0.67

(Jim, Mary) = 0.75

Reflections:

Reflecting on this analysis highlights a key aspect of data preprocessing that context dictates methodology. In medical datasets like this one, treating variables symmetrically can result in skewed conclusions. If standard matching coefficients that include negative matches (0) was used, two healthy individuals who tested negative for every disease would be flagged as highly similar. However in medicine, the absence of a symptom is the norm. The shared presence of specific positive test results or symptoms is what reveals a meaningful epidemiological connection or shared diagnosis.

Furthermore, the process underscores how a simple distance metric can translate raw data into actionable insights. The calculations immediately separate the individuals into distinct profiles: Jack and Mary exhibit a high level of similarity, 0.67, pointing to a shared underlying pathology since both tested positive for **Test-1** alongside a fever. Jim stands completely isolated with the highest dissimilarity scores. Because his profile is driven by an entirely different clinical presentation, a cough instead of positive tests, a medical professional looking at these results could quickly triage Jim into a different diagnostic or treatment pathway than Jack and Mary.